The Journal of Open Source Software

Surrogate endpoint validation using the R package frailtypack

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DOI: 10.xxxxxx/draft

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Editor:

Submitted: 20 August 2025
Published: unpublished

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In this article we present two functions of the `frailtypack` package for investigating surrogate endpoints through mediation analysis. A detailed explanation of these functions can be found in the reference manual of the package available in R or on the CRAN (Rondeau & Gonzalez, 2005). Results in this article were obtained using `frailtypack` version 3.7.0 and R version 4.4.0.

Function jointSurroPenal

The function `jointSurroPenal` investigates surrogacy when both the surrogate endpoint and the final endpoint are time-to-event. The call to this function is as follows,

```
R> model<- jointSurroPenal(data, maxit = 50,
+          indicator.zeta = 1, indicator.alpha = 1,
+          frail.base = 1, n.knots = 6, LIMparam = 0.001,
+          LIMlogl = 0.001, LIMderiv = 0.001, nb.mc = 300,
+          nb.gh = 32, nb.gh2 = 20, adaptatif = 0, int.method = 2,
+          nb.iterPGH = 5, nb.MC.kendall = 10000, nboot.kendall = 1000,
+          true.init.val = 0, theta.init = 1, sigma.ss.init = 0.5,
+          sigma.tt.init = 0.5, sigma.st.init = 0.48,
+          gamma.init = 0.5, alpha.init = 1,
+          zeta.init = 1, betas.init = 0.5, betat.init = 0.5,
+          scale = 1, random.generator = 1, kappa.use = 4,
+          random = 0, random.nb.sim = 0,
+          seed = 0, init.kappa = NULL,
+          ckappa = c(0,0), nb.decimal = 4,
+          print.times = TRUE, print.iter = FALSE,
+          mediation=FALSE, g.nknots=1,
+          pte.times=NULL, pte.ntimes=NULL,
+          pte.nmc=500, pte.boot=FALSE, pte.nboot=2000,
+          pte.boot.nmc=500, pte.integ.type=2)
```

In order to use this function, the user must provide a dataset (argument `data`) with the following structure:

```
R> head(data)

patientID    timeT    timeS statusT statusS trt trialID
        1 9.057946 2.217739       1       1   0       1
        2 2.986813 1.389263       1       1   0       1
        3 8.874237 8.874237       1       0   1       1
        4 3.245388 1.809671       1       1   1       1
        5 4.448964 2.603604       1       1   0       1
```

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The dataset must contain one line per subject and seven columns: one for the subject's identification number (column `patientID`), for the trial number (`trialID`), treatment indicator (`trt`), for the follow-up time for the surrogate (`timeS`) and censoring indicator (`statusS`) and for the follow-up time for the final endpoint (`timeT`) and censoring indicator (`statusT`).

The option to investigate surrogacy through mediation analysis is given by setting the argument `mediation` to `TRUE`. In that case, a function $\gamma(S)$ is estimated using a basis of B-splines whose number of knots is given by the argument `g.nknots`, which can take any value between 1 and 5 and the time-dependent proportion of treatment effect, $\mathrm{PTE}(t)$, will be estimated. The timepoints at which this function has to be evaluated can be specified through the argument `pte.times`. If one does not want to specify any timepoints, the argument `pte.ntimes` can be used instead to specify the number of timepoints at which $\mathrm{PTE}(t)$ should be evaluated. These points will then be selected evenly on the range of the observed event times. The argument `pte.boot` is used if we want to compute quantile-based confidence bands of $\hat{\mathrm{PTE}}(t)$ using parametric bootstrap. If set to `TRUE`, then the number of bootstrap samples to be used can be set with `pte.nboot`.

A complete description of all parameters can be found in the documentation of the function in the package.

The function `jointSurroPenal` returns an `R` object of class `jointSurroPenal` if the argument `mediation` is set to `FALSE` and of class `jointSurroMed` otherwise. In both cases, common `R` functions such as `summary`, `print` and `plot` can be to display the results.

Function longiPenal

The function `longiPenal` can be used to investigate surrogacy when the surrogate outcome is a longitudinal biomarker and the final endpoint is a time-to-event. The call to this function is as follows (the values given to each parameters are the default values):

```
R> model<- longiPenal(formula, formula.LongitudinalData,
+           data,  data.Longi, formula.Binary = FALSE,
+           random, random.Binary = FALSE,
+           fixed.Binary = FALSE, GLMlog = FALSE,
+           MTP = FALSE, id, intercept = TRUE,
+           link = "Random-effects", timevar = FALSE,
+           left.censoring = FALSE, n.knots,
+           kappa, maxit = 350,  hazard = "Splines",
+           mediation = FALSE, med.center = NULL, med.trt = NULL,
+           init.B, init.Random, init.Eta,
+           method.GH = "Standard", seed.MC = 1, n.nodes,
+           LIMparam = 1e-3, LIMlogl = 1e-3, LIMderiv = 1e-3,
+           print.times = TRUE,  med.nmc = 500, pte.times = NULL,
+           pte.ntimes = NULL, pte.nmc = 500,
+           pte.boot=FALSE,pte.nboot=2000)
```

This function requires the specification of two datasets. The first one, specified through the argument `data`, contains the data regarding the final endpoint including the follow-up time for each subject, the censoring indicator, and potential covariates. Note that this dataset requires one line per subject and therefore does not allow for time-dependent covariates to be included. Associated with this dataset is the `formula` object, with the response on the left of a $\sim$ operator, and the covariates on the right. The response must be a survival object as returned by the `Surv` function of the R `survival` package (Therneau, 2024). The variables used in `formula` should be the ones contained in `data`. For the longitudinal part, the repeated measurements data are specified in a separate dataset through the argument `data.Longi`. The specification of the longitudinal submodel is made through `formula.LongitudinalData` which is a `R` formula with the observed biomarker on the left and the different covariates on the

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right. Both the names for the biomarker and the covariates specified in this formula should correspond to columns in the dataset `data.Longi`.

Both `data` and `data.Longi` should have a column labelled "`id`" that corresponds to the identificator of each subject in order to link the two datasets, i.e., `id=1` in `data` should corresponds to the same individual with `id=1` in `data.Longi`. Note that for simpliciy the variable `id` should takes values between 1 and n where n is the total number of subjects.

The mediation analysis is enabled by setting the argument `mediation` to `TRUE`. In that case one should also specify the name of the variable in `data` that corresponds to the treatment through the argument `treatment`. If `mediation` is set to `TRUE` then the function $\mathrm{PTE}(t)$ will be estimated. As for the function `jointSurroPenal`, one can specify the timepoints at which $\hat{\mathrm{PTE}}(t)$ should be evaluated or the number of timepoints at which it should be evaluated. The argument `pte.boot` takes values `TRUE/FALSE` to indicate if the parametric bootstrap estimation of the standard-error of $\hat{\mathrm{PTE}}(t)$ and its confidence bands should be computed. If set to `TRUE` then the number of bootstrap samples is specified by `pte.nboot`. A complete description of all parameters can be found in the documentation of the function in the package.

The function `longiPenal` returns a R object of class `longiPenal` on which the usual R functions `summary`, `print` and `plot` can be applied as will be illustrated in Section 4.

Illustrations

We illustrate the two functions in two applications on cancer data from meta-analyses or multicentric randomized clinical trial. The first application is based on a dataset on gastric cancer and the second on colorectal cancer. In the following we assume that the `frailtypack` package is loaded using the R commands `require(frailtypack)` or `library(frailtypack)`.

Time-to-relapse as a surrogate of overall survival using proportion of treatment effect in gastric cancer: a mediation approach

The first application is on a meta-analysis on resectable gastric cancer patients investigating the addition of adjuvant chemotherapy after surgery versus surgery alone (Paoletti et al., 2010). In this illustration, the final endpoint is the time between randomization and death from any cause while the surrogate is the time-to-relapse, defined as the time between randomization and disease recurrence or occurrence of a second cancer, whichever occurred first. Therefore both endpoints might be right censored due to loss to follow-up and moreover the surrogate endpoint might be censored by the final endpoint. We are interested in estimating the proportion of treatment effect (adjuvant chemotherapy or not) on overall survival that goes through its effect on time-to-relapse.

Dataset

The dataset `gastadj` can be loaded directly from `frailtypack` using the command

```
R>  data("gastadj")
```

This dataset contains the data of 3288 patients from 14 randomized clinical trials. Out of these 3288 patients, 1654 were assigned to the control group of no adjuvant chemotherapy, and the remaining 1634 patients were assigned to receive adjuvant chemotherapy. The dataset has the following structure, using the command `head(gastadj)`,

```
R>  head(gastadj)

trialID patientID trt timeT statusT timeS statusS
1       1         1   1  4636       0  4636       0
2       1         2   1  4536       0  4536       0
3       1         3   0  3151       1  3151       1
```

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```
4        1          4   1   485        1   432        1
5        1          5   0   435        1   300        1
6        1          6   0   187        1   137        1
```

The columns `trialID`, `patientID`, `trt` are the trial, patient and treatment indicator respectively. The variables `timeT`, `timeS` are the follow-up times for the final endpoint and surrogate endpoints respectively and `statusT`, `statusS` their associated censoring indicator. In this dataset, the variable `timeS` corresponds to a time-to-progression defined as the earliest between cancer recurrence, occurrence of a second cancer or death. Therefore this endpoint includes death as a composite endpoint which raises questions from a mediation analysis viewpoint since the final endpoint always triggers the surrogate. To circumvent this, we instead analyzed the time-to-relapse (cancer recurrence or second cancer) by censoring them at the time of death. In the dataset this change can be made using the following command,

```
R>  gastadj[gastadj$timeS == gastadj$timeT &
+   gastadj$statusS == 1, c("statusS")] <- 0
```

For practical purposes, and to reduce the computation time, we restrain this illustration on a subset of the original dataset, by selecting 20% of the patients at random.

Moreover, to circumvent some computing issues, we divide the time variable (originally represented in a daily scale) by 365 in the yearly scale. Therefore, the full call for data preparation is

```
R>  data(gastadj)
R>  gastadj$timeS <- gastadj$timeS/365
R>  gastadj$timeT <- gastadj$timeT/365
R>  #"statusS" corresponds now
R>  #to a time-to-relapse event
R>  gastadj[gastadj$timeS == gastadj$timeT &
R>  gastadj$statusS == 1, c("statusS")] <- 0
R>  # select 20% of the original dataset
R>  set.seed(1)
R>  n <- nrow(gastadj)
R>  subset <- gastadj[sort(sample(1:nrow(gastadj),
+              round(n*0.2), replace = F)),]
```

Model fitting and surrogacy evaluation

The call to the function `jointSurroPenal` is the following:

```
R>  mod.gast<-jointSurroPenal(subset,n.knots = 4,
+              indicator.zeta = 0, indicator.alpha = 0,
+              mediation = TRUE, g.nknots = 1,
+              pte.times = seq(1.5, 2,length.out = 30),
+              pte.nmc = 10000, pte.boot = TRUE, pte.nboot = 1000,
+              pte.boot.nmc = 1000)
```

Here we specify that `mediation = TRUE`, therefore the function $\gamma(S)$ will be estimated. The number of inner knots used in the spline basis is fixed to 1 via the command `g.nknots=1`. Since we are interested in the mediation analysis setting, we specify that we want the function $\mathrm{PTE}(t)$ to be evaluated at 30 timepoints defined by the argument `pte.times`. The number of Monte-Carlo points used in the approximation of the integral over the random effects is set to 10000. The use of parametric bootstrap to derive estimated standard-errors and confidence bands for $\mathrm{PTE}(t)$ is given by `pte.boot=TRUE` where we also specify that we want this bootstrap to be based on 1000 sampling via `pte.nboot=1000`. Finally, for illustration purposes and to reduce computation time, we also set the number of Monte Carlo points used for each bootstrap sample to 1000. However, in practice this number should be the same as

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for the estimation of $\mathrm{PTE}(t)$. The object `mod.gast` has R class `jointSurroMed`, and we can apply the `summary` function to display the results.

```
R>  summary(mod.gast)
R>  plot(mod.gast, type.plot = "Hazard", plot.mediation= "All")
```

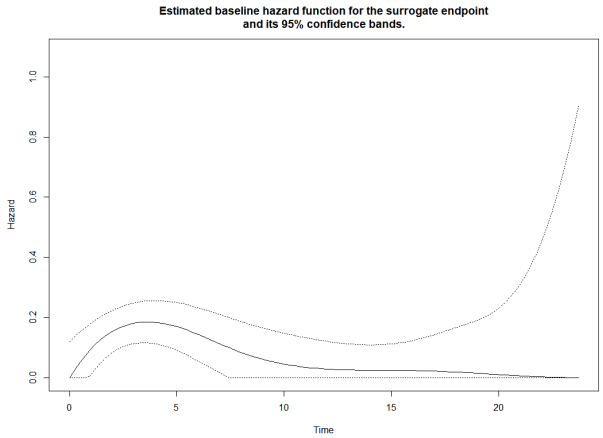

Figure 1: image

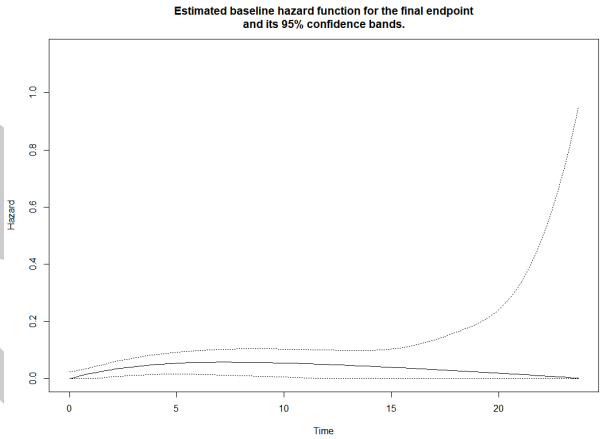

Figure 2: image

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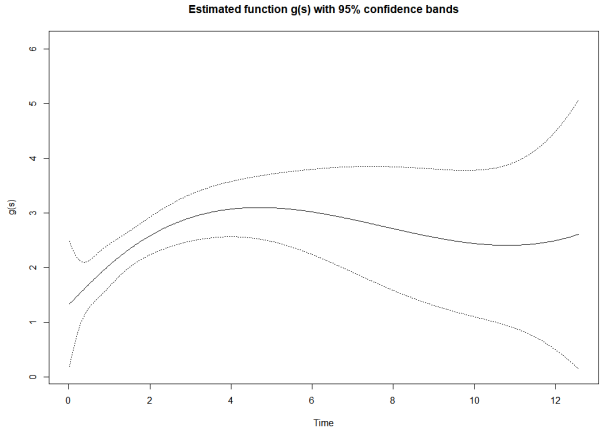

Figure 3: image

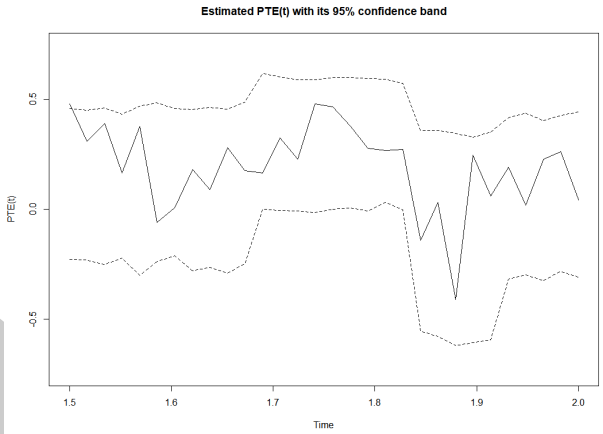

Figure 4: image

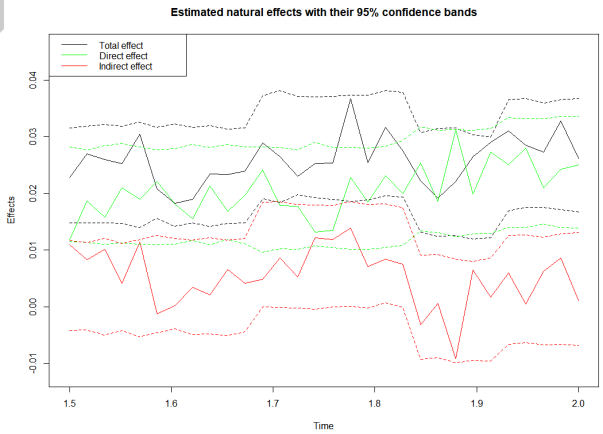

Figure 5: image

Tumor size as a surrogate biomarker of overall survival in colorectal cancer: a mediation approach

In this second application we are interested in evaluating the tumor size evolution over time as a surrogate of the overall surival in colorectal cancer. Since the tumor size evolution is a longitudinal biomarker we will base the analysis on the function `longiPenal`. We will use a dataset containing 150 patients randomly selected from the FFCD 2000-05 multicenter phase III clinical trial (Ducreux et al., 2011). This trial originally included 410 patients with metastatic colorectal cancer randomized into two treatment strategies: combination and sequential chemotherapy. The dataset contains times of observed appearances of new lesions censored by a terminal event (death) with some baseline characteristics. Because the available dataset does not contain the identificator of the center of the patients and for computational purposes, we illustrate the approach without taking into account the multi-centric nature of the data. The data are actually composed of two datasets, one for the survival part and another containing the repeated measurements of tumor sizes.

Dataset

As for the two previous illustrations these two datasets can be loaded from `frailtypack`.

```
R>  data(colorectal)
R>  data(colorectalLongi, package = "frailtypack")
```

The dataset `colorectal` contains several observations per subject, one for each new lesions in addition to a follow-up time and a censoring indicator for death. Therefore we only want to retrieve the last observation for a subject. In this dataset the variable `new.lesions` takes the value 1 if a new lesion is record and 0 otherwise. Therefore if a subject has n_i observations, the observations $1, \ldots, n_{i-1}$ all have the status `new.lesions` equals to 1 (since the repeated follow-up are based on the appearance of new lesions). Hence, the last observation for each subject can be taken as the only one for which `new.lesions` equals 0:

```
R>  colorectalSurv <- subset(colorectal, new.lesions == 0)
```

In the dataset the variable `treatment` takes the value `"S"` for "sequential" and `"C"` for "combined", for interpretability we simply make this variable binary $0/1$,

```
R>  colorectalSurv$treatment <- sapply(colorectalSurv$treatment,
+   function(t) ifelse(t == "S", 1, 0))
R>  colorectalLongi$treatment <- sapply(colorectalLongi$treatment,
+   function(t) ifelse(t == "S", 1, 0))
```

To keep the illustration simple we only adjust on the variable age as a categorical variable: $<$60 years, 60-69 years or $>$69 years.

Model fitting and surrogacy evaluation

The call to the function is:

```
R>  mod.col = longiPenal(Surv(time1, state) ~ age + treatment,
+             tumor.size ~ age + year*treatment,
+             data = colorectalSurv, data.Longi = colorectalLongi,
+             random = c("1", "year"), id = "id",
+             link = "Current-level", timevar = "year",
+             method.GH = "Pseudo-adaptive",
+             mediation = TRUE,
+             med.trt = colorectalSurv$treatment,
+             med.center = NULL,
+             n.knots = 7, kappa = 2,
+             pte.times = seq(1,2,length.out = 30),
```

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```
+             pte.boot = TRUE, pte.nboot = 2000,
+             pte.nmc = 1000)
```

In this call we fit a model using a "Current-level" link function between the longitudinal biomarker and the final endpoint. We specify a random slope and intercept in the longitudinal submodel. The arguments `n.knots` and kappa specify the number of knots and the penalization term related to the splines baseline hazard function> The argument `mediation = TRUE` indicates that we want to compute the natural direct and indirect effects as well as the proportion of treatment effect, $\mathrm{PTE}(t)$. We require that this function to be evaluated at 30 timepoints between 1 and 2 through the argument `pte.times`. Moreover, we also require that the bootstrap standard error and confidence interval for $\mathrm{PTE}(t)$ computed using 2000 samples. Finally, `pte.nmc` specify the number of Monte Carlo sample to be used for integrating over the random effects distributions for the computation of the mediation-related quantities such as the $\mathrm{PTE}(t)$ and the natural direct and indirect effects. The result can be displayed by applying the R function `print` to the object `mod.col`.

```
R>  print(mod.col)
```

The estimated baseline hazard function $\hat{\lambda}_{0,T}(t)$, $PTE(t)$ and estimated natural effects can be plotted using the R function `plot`.

```
R>  plot(mod.col, plot.mediation = "All", conf.bands = TRUE)
```

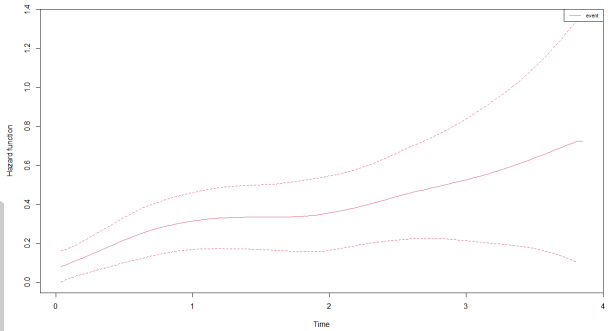

Figure 6: image

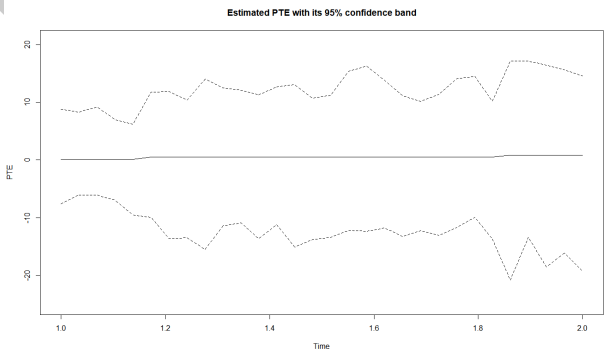

Figure 7: image

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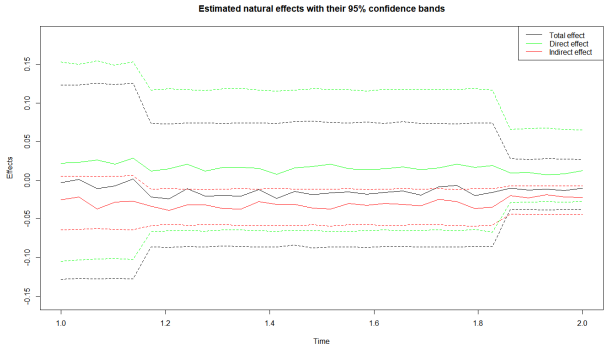

Figure 8: image

Discussion

Further developments of the `frailtypack` package will concern the extension of the proposed functions to validate surrogate endpoints. In order to improve the flexibility of the proposed approaches, other options for the type of surrogate or final endpoint will be proposed, for example in the case of a binary final endpoint.

Acknowledgements

Q. Le Coënt and C. Legrand acknowledge the support of the ARC project IMAL (grant 20/25-107) financed by the Wallonia-Brussels Federation and granted by the Académie universitaire Louvain.

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